ME4640 HW 6 (FK setup, Manipulator Jacobian)

Monday, April 20, 2026 8:03 AM

ME4640
HW 6 (FK ...

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HW #6: WRITTEN

Box answers and show all work in the space provided. Submit at the start of class on the scheduled due date.

For the following questions, consider the manipulator shown in Figure 1. The first axis is a rotation about the z-axis of the spatial frame. The second axis is a rotation about the y-axis of the spatial frame as shown.

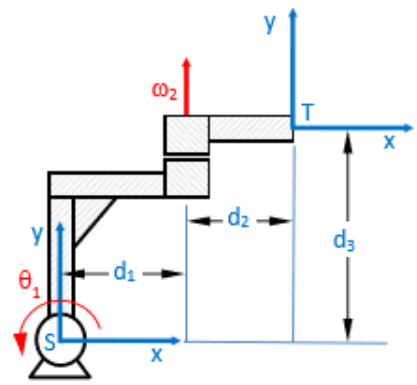

Figure 1. A 2 DOF manipulator shown in the home configuration.

1) Fine each joint axis for FK calculations.
   Use ω for rotations and v for translations.

$$\omega_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \omega_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

2) Find a point on each joint axis, $\mathbf{q}_{1-2}$. Indicated not-needed as necessary.

$$q_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad q_2 = \begin{bmatrix} d_1 \\ 0 \\ 0 \end{bmatrix}$$

Not Needed

3) Find the twists, ξ_{1-2}, necessary for kinematic analysis.

$$v_1 = -\hat{\omega}_1 q_1 = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$v_2 = -\hat{\omega}_2 q_2 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} d_1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ d_1 \end{bmatrix}$$

$$\xi_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \qquad \xi_2 = \begin{bmatrix} 0 \\ 0 \\ d_1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

4) Find the home position, $\mathbf{g}_{st}(0)$, of the tool frame.

$$R_{st} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$g_{st}(0) = \begin{bmatrix} 1 & 0 & 0 & d_1 + d_2 \\ 0 & 1 & 0 & d_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$P_{st} = \begin{bmatrix} d_1 + d_2 \\ d_3 \\ 0 \end{bmatrix}$$

5) Find the spatial manipulator Jacobian, $\mathbf{J}^s_{st}(\theta)$. You can confirm via Matlab but please show the calculations steps.

$$J^s_{st}(\theta) = \begin{bmatrix} \xi_1 & , & Ad(e^{\xi_1\theta_1})\xi_2 \end{bmatrix}$$

$$\xi_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad \hat{p} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Ad(e^{\xi_1\theta_1})\xi_2 \begin{bmatrix} c_1 & -s_1 & 0 & 0 & 0 & 0 \\ s_1 & c_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & c_1 & -s_1 & 0 \\ 0 & 0 & 0 & s_1 & c_1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ d_1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ d_1 \\ -s_1 \\ c_1 \\ 0 \end{bmatrix}$$

$$J^s_{st}(\theta) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & d_1 \\ 0 & -s_1 \\ 0 & c_1 \\ 1 & 0 \end{bmatrix}$$

$$e^{\hat{\xi}_1\theta_1} = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$e^{\hat{\xi}_2\theta_2} = \begin{bmatrix} c_2 & 0 & s_2 & d_1(1-c_2) \\ 0 & 1 & 0 & 0 \\ -s_2 & 0 & c_2 & d_1 s_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{p}_{st}(\theta) = \begin{bmatrix} d_1c_1 - d_3s_1 + d_2c_1c_2 \\ d_3c_1 + d_1s_1 + d_2c_2s_1 \\ -d_2s_2 \end{bmatrix}$$